



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 • STANDARD 6.NOS.1

Fraction Division Problem Solving

Lesson 6-11 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can solve real-world problems by writing and solving fraction division equations.

Language Objective: I can explain my reasoning using the words model, equation, solution, and reasonableness.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Fraction Division Problem Solving

Model

Equation

Solution

Reasonableness

Inverse operations



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

Name the total and the size of each group, then divide the

by the .

2

A drawing or diagram that shows a math situation is a

.

3

A math sentence that says two amounts are equal, with an = sign, is an

.

4

The value that makes an equation true is the .

- 5 Checking whether an answer makes sense for the problem is checking its

.

- 6 Operations that undo each other, like multiplication and division, are

.

Write About the Math

How does setting up a fraction division equation help solve this real-world problem?

¿Cómo ayuda armar una ecuación de división de fracciones a resolver este problema del mundo real?

Say your answer to a partner first. Then write it, using at least two of these words:

☐

Fraction Division Problem Solving

Resolución de problemas de división de fracciones

☐

Model

Modelo

☐

Equation

Ecuación

☐

Solution

Solución

☐

Reasonableness

Razonabilidad

Model: *The total amount is the dividend; the size of each test is the divisor.* — Students identify $2/3$ as the total being split (dividend) and $1/6$ as the group size (divisor), writing $2/3 \div 1/6$.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The chemist can run ____ tests because $5/6 \div 1/12 =$ ____.**
- **My answer is ____ because ____.**

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
- **My evidence is ____, so ____.**
Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Fraction Division Problem Solving
2. **Notes 2:** Model
3. **Notes 3:** Equation
4. **Notes 4:** Solution
5. **Notes 5:** Reasonableness
6. **Notes 6:** Inverse operations
7. **Watch for this mistake:** *A common mistake in Fraction Division Problem Solving is reversing which fraction is the dividend and which is the divisor when setting up the equation from a word problem. For example: a pitcher holds $\frac{3}{4}$ cup of punch, and each serving is $\frac{1}{8}$ cup — how many servings? The correct setup is $\text{total} \div \text{group size} = \frac{3}{4} \div \frac{1}{8} = \frac{3}{4} \times \frac{8}{1} = \frac{24}{4} = 6$ servings. A student who reverses the setup calculates $\frac{1}{8} \div \frac{3}{4} = \frac{1}{8} \times \frac{4}{3} = \frac{4}{24} = \frac{1}{6}$, an answer smaller than 1 that can't possibly be a serving count. Before you submit, check: does my answer make sense as a NUMBER OF GROUPS? If the setup gives a fraction less than 1 for a 'how many' question, you likely swapped the dividend and divisor.*
8. **Try It 1:** A. 5 — *The total distance ($\frac{5}{6}$ mile) is the dividend and the checkpoint spacing ($\frac{1}{6}$ mile) is the divisor: $\frac{5}{6} \div \frac{1}{6} = \frac{5}{6} \times \frac{6}{1} = \frac{30}{6} = 5$ checkpoints.*
9. **Try It 2:** A. 3 — *The total ($\frac{9}{10}$ liter) is the dividend and each vial ($\frac{3}{10}$ liter) is the divisor: $\frac{9}{10} \div \frac{3}{10} = \frac{9}{10} \times \frac{10}{3} = \frac{90}{30} = 3$ vials, so the code is 3.*